Quantum Speedups in Regret Analysis of Infinite Horizon Average-Reward Markov Decision Processes

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Problem Setup

Motivation: Average reward reinforcement learning have many crucial applications since it targets the steady-state and long-term performance, yet classical algorithms under this setting have a $\tilde{O}(\sqrt{T})$ regret lower bound.

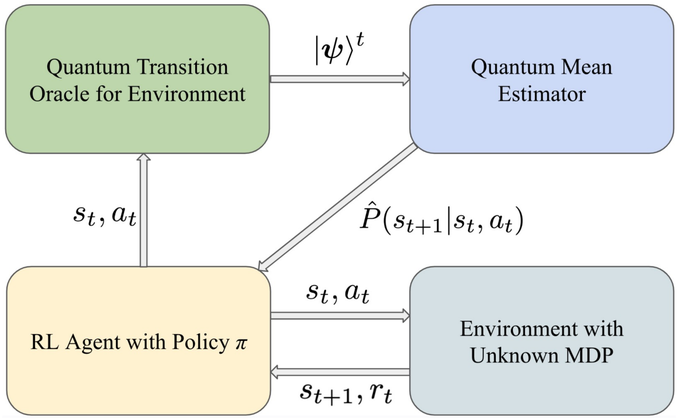
Key question: Can quantum computing provably break this barrier of a $\tilde{O}(\sqrt{T})$ regret in average reward RL?.

Setting:

- Infinite-horizon average reward MDP
- Access to the quantum transition oracle

$$\mathcal{U}_P : |s_t\rangle \otimes |a_t\rangle \otimes |0\rangle \to |\psi\rangle^{\mathbf{t}}$$
$$|\psi\rangle^{\mathbf{t}} = |\mathbf{s_t}\rangle \otimes |\mathbf{a_t}\rangle \otimes \textstyle\sum_{\mathbf{s'}\in\mathcal{S}} \sqrt{\mathbf{P}(\mathbf{s'}|\mathbf{s_t},\mathbf{a_t})}|\mathbf{s'}\rangle$$

Proposed framework:



Goal: Maximize the average reward with a smaller regret fewer than any classical methods.

Challenge

Only one measurement per quantum sample ⇒ must reuse information without repeated queries.

Key Methodology

Epoch-based optimistic planning: at each epoch e solve:

$$\max_{\{\rho(s,a),P_e(\cdot|s,a)\}_{(s,a)\in\mathcal{S}\times\mathcal{A}}} \sum_{(s,a)\in\mathcal{S}\times\mathcal{A}} r(s,a)\rho(s,a)$$
$$\sum_{s,a}\rho(s,a)=1, \rho(s,a)\geq 0 \quad \forall (s,a)\in\mathcal{S}\times\mathcal{A}$$
$$\sum_{s'\in\mathcal{S}} P_e(s'|s,a)=1, P_e(\cdot|s,a)\geq 0 \ \forall (s,a)\in\mathcal{S}\times\mathcal{A}$$
$$\sum_{a\in\mathcal{A}}\rho(s',a) = \sum_{(s,a)\in\mathcal{S}\times\mathcal{A}} P_e(s'|s,a)\rho(s,a), \forall s'\in\mathcal{S}$$
$$\|P_e(\cdot|s,a)-\hat{P}_e(\cdot|s,a)\|_1 \leq \frac{7S\log(S^2At_e)}{\max\{1,N_e(s,a)\}}$$

Obtain quantum samples of counters in each epoch e

$$\{|\psi\rangle\}_i^e = \{|\psi\rangle^t \mid t_{e-1}^{start} \leq t \leq t_e^{start}-1, (s_t,a_t)=(s_i,a_i)\}$$

Perform quantum mean estimation in the end of each epoch, using each quantum samples only a single time.

$$\tilde{P}_e(\cdot|s_i,a_i) = \texttt{QBounded}_S(\{P_e(\cdot|s_i,a_i), L_2, n_e^{s_i,a_i}, \delta_e)$$

A momentum-fusion procedure aggregates mean estimates from every epoch, reusing each once-measured quantum sample exactly once while preserving the original accuracy order.

$$\hat{P}_e(\cdot|s_i,a_i) = \begin{cases} 0 & \text{if } e=1 \\ \tilde{P}_e(\cdot|s_i,a_i) & \text{if } e=2 \\ \hat{w}\hat{P}_{e-1}(\cdot|s_i,a_i)+\tilde{w}\tilde{P}_e(\cdot|s_i,a_i) & \text{if } e\geq 3 \end{cases}$$

$$\hat{w} = \frac{N_{e-1}(s_i,a_i)}{N_{e-1}(s_i,a_i)+\nu_{e-1}(s_i,a_i)} \qquad \tilde{w} = \frac{\nu_{e-1}(s_i,a_i)}{N_{e-1}(s_i,a_i)+\nu_{e-1}(s_i,a_i)}$$

Update policy in the start of each epoch

$$\pi_e(a|s) = \frac{\rho_e(s,a)}{\sum_{\tilde{a}\in\mathcal{A}}\rho_e(s,\tilde{a})}$$

Results

Assumptions:

- Finite mixing time
- Deterministic and known reward

Quantum speedup for mean estimation in [1]

Lemma 1 (Quantum multivariate bounded estimator, Theorem 3.3 of (Cornelissen et al., 2022)). *Let X be a d-dimensional bounded random variable such that $||X||_2 \leq 1$. Given three reals $L_2 \in (0,1]$, $\delta \in (0,1)$ and $n \geq 1$ such that $\mathbb{E}[||X||_2] \leq L_2$, the multivariate bounded estimator $\texttt{QBounded}_d(X, L_2, n, \delta)$ obtained by Algorithm 1 performs $f(n,d) = \mathcal{O}\left(n\log^{1/2}(n\sqrt{d})\right)$ quantum experiments and outputs a mean estimate $\hat{\mu}$ of $\mu = \mathbb{E}[X]$ such that,*

$$\mathbb{P}\left[||\hat{\mu}-\mu||_\infty \leq \frac{\sqrt{L_2}\log(d/\delta)}{n}\right] \geq 1-\delta. \qquad (1)$$

Main theorem: Logarithmic regret

$$\mathcal{R}_{[0,T-1]} = \mathcal{O}\left(S^5A^4T_{mix}\log^3\left(\tfrac{T}{SA}\right)\log^{\frac{1}{2}}(T\sqrt{S})\log(S^2AT)\right)$$

Contributions

- First quantum algorithm with provable guarantees for average reward reinforcement learning.
- Coherent embedding of model-based pipeline into a quantum framework.
- Obtained logarithmic regret speedup $\tilde{O}(1)$ compared with the classical result $\tilde{O}(\sqrt{T})$.

References

[1] Cornelissen et al, Near-optimal quantum algorithms for multivariate mean estimation, STOC 2022